

# PMx Weekly Digest: May 26, 2026

Content Source: Full Text

## Paper 1: Assessment and Benchmarking of Model Informed Approaches in Drug Development for

### Bibliographic Information

- **Title:** Assessment and Benchmarking of Model Informed Approaches in Drug Development for Hemoglobinopathies: A Review of Scientific Advices From January 2000 to December 2024
- **Authors:** Govere GS, Dogné JM, Musuamba F
- **Journal:** N/A (preprint/presentation at PAGE 2025)
- **DOI:** <https://pubmed.ncbi.nlm.nih.gov/41733111/>

**Paper Type:** GENERIC

### Quick Take

This review analyzes EMA scientific advices for sickle cell disease and thalassemia from 2000–2024 to assess the use of modeling and simulation (M&S) in drug development. Using a question-centric approach, the authors built a repository of regulatory questions and benchmarked M&S methods against alternative approaches. The study finds that M&S is underutilized and highlights the need for structured frameworks like ICH M15 to improve regulatory acceptance.

---

### Executive Summary

The authors systematically reviewed 45 and 31 scientific advices (SAs) for sickle cell disease (SCD) and thalassemia, respectively, from the EMA between 2000 and 2024. They translated sponsor questions into regulatory questions using a question-centric approach, classifying them by benefit–risk, efficacy, safety, and pharmacokinetics at multiple biological levels. Only 33% of SCD SAs and 20% of thalassemia SAs mentioned M&S, with population PK and exposure–response models being most common, primarily used in Phase 2/3. The ICH M15 credibility assessment framework was applied to evaluate model acceptance criteria. Benchmarking identified 14 questions where M&S could replace traditional methods (e.g., standard statistical analysis, NOAEL). The work underscores the need for transparent, standardized criteria for M&S acceptance in regulatory submissions.

---

### Scientific Context & Motivation

Drug development for hemoglobinopathies (SCD and thalassemia) faces challenges including small patient populations, pediatric involvement, and limited curative options. Modeling and simulation (M&S) offer potential to improve efficiency through data integration, extrapolation, and predictive modeling, but their regulatory acceptance remains inconsistent. This study addresses the gap by systematically reviewing EMA scientific advices to understand how M&S has been proposed and assessed, and to identify opportunities for broader adoption using the ICH M15 framework.

---

## ✂ Methodological Snapshot

The authors retrieved 45 SCD and 31 thalassemia scientific advices from EMA’s IRIS platform using Scientific Explorer. They extracted and anonymized sponsor questions, then reformulated them into regulatory questions using a question-centric approach. Questions were classified by benefit–risk, efficacy, safety, and PK at molecular, cellular, tissular, individual, and population levels. M&S methods were identified via keyword search and categorized by type (POPPK, E-R, PBPK, MMRM, etc.) and development stage. The ICH M15 credibility assessment framework was applied to evaluate regulatory comments on M&S. Benchmarking was performed for questions where both M&S and alternative methods were proposed, comparing model impact and acceptance criteria.

---

## 📋 Detailed Analysis

The study’s strength lies in its systematic, structured approach to analyzing regulatory interactions. The question-centric repository reveals that most regulatory questions are at the population level for efficacy and safety, reflecting late-stage development. M&S is concentrated in PK characterization and dose selection, with POPPK and E-R models being workhorses. The benchmarking exercise (Table 2) is particularly valuable: it shows that M&S can address high-impact questions (e.g., indication extension) with appropriate credibility assessment. However, the low overall M&S usage (5 – 6% of questions) suggests persistent barriers, possibly including lack of familiarity, data requirements, or unclear acceptance criteria. The ICH M15 framework offers a path forward by making model risk and impact explicit, but its adoption requires cultural change and training. The authors’ call for an openly accessible structured framework is well-founded; the ERAMET project’s ambition to link model risk/impact to quantitative acceptance criteria could be transformative.

---

## Domain Context

Hemoglobinopathies are rare diseases with high unmet need, making efficient drug development critical. Regulatory flexibility is often sought through scientific advice. This paper sits at the intersection of regulatory science, pharmacometrics, and rare disease drug development. The ICH M15 guideline (finalized in 2024) provides a global framework for MIDD credibility assessment, and this study is among the first to apply it retrospectively to real regulatory data. The findings are relevant to sponsors developing drugs for SCD/thalassemia and to regulators evaluating M&S evidence in rare diseases.

---

## 📌 Key Findings

1. M&S was mentioned in only 33% of SCD and 20% of thalassemia SAs, indicating underutilization.
2. Population PK and exposure–response models were the most common M&S methods, used mainly in Phase 2/3 studies.
3. M&S addressed ~ 6% of all regulatory questions, predominantly at individual/population levels for efficacy and safety.
4. Benchmarking identified 14

questions (e.g., dose selection, extrapolation, endpoint adequacy) where M&S could replace or complement traditional methods. 5. The ICH M15 framework provides a structured way to assess model credibility and communicate regulatory expectations.

---

## Strengths & Limitations

### Strengths

- Systematic review of a large set of EMA scientific advices over 24 years
- Novel application of the question-centric approach to build a structured repository
- Explicit linkage of M&S methods to regulatory questions and ICH M15 criteria
- Benchmarking of M&S against alternative methods provides practical guidance

### Limitations (Acknowledged by Authors)

- Confidentiality constraints required anonymization, limiting granularity
- Small number of SAs mentioning M&S (15 for SCD, 6 for thalassemia)
- Potential selection bias as SAs are voluntary and may not represent all development programs

### Limitations (Expert Review)

- No quantitative analysis of model performance or acceptance rates
- Lack of validation of the repository structure across other disease areas
- Reliance on keyword search may miss implicit M&S use

### Generalizability

The findings are specific to hemoglobinopathies and EMA regulatory context; however, the question-centric approach and ICH M15 framework are broadly applicable to other rare diseases and regulatory settings.

---

---

## Figures & Tables

- **Figure 1:** Schematic of the process to compile the repository of regulatory questions, methods, and data.
  - *Significance:* Illustrates the workflow from SA retrieval to repository building and ICH M15 implementation.
- **Figure 2:** Steps to anonymize and reformulate applicant questions into repository questions.
  - *Significance:* Shows the transformation from sponsor-specific to regulatory-generic questions.
- **Figure 3:** Overall structure of the repository with benefit–risk at the center, supported by efficacy, safety, and PK layers.
  - *Significance:* Demonstrates the hierarchical, question-centric organization that facilitates benchmarking.
- **Figure 4:** Number of translated questions per benefit–risk classification and hierarchical level, with M&S proportions.
  - *Significance:* Quantifies the distribution of questions and highlights the low M&S usage.

- **Figure 5:** Example of ICH M15 credibility assessment for a model used in efficacy extrapolation, benchmarked against standard statistical analysis.
    - *Significance:* Concrete illustration of how the framework enables transparent comparison of methods.
  - **Table 1:** Use of M&S methods across drug development stages for SCD and thalassemia.
    - *Significance:* Shows that M&S is most used in Phase 2/3, with POPPK and E-R being dominant.
  - **Table 2:** Benchmarking of M&S models against alternative methods for 14 regulatory questions.
    - *Significance:* Provides a practical list of questions where M&S can be considered as an alternative, with model impact levels.
- 

### Supplementary Materials

Table S1 correlates CHMP comments on M&S methods with ICH M15 key assessment elements, including model impact assessment.

---

### Future Directions

Expand the repository to other rare diseases and regulatory agencies; develop quantitative criteria linking model risk and impact to acceptance thresholds; prospectively evaluate the ICH M15 framework in ongoing drug development programs.

---

### Expert Commentary

This work is a timely and practical contribution to the MIDD field. By systematically mapping regulatory questions to M&S methods and benchmarking them against traditional approaches, the authors provide a blueprint for operationalizing the ICH M15 guideline. The repository could serve as a living resource for sponsors and regulators to align expectations and accelerate the adoption of model-informed evidence in rare disease drug development.

---

### Bottom Line

M&S is underused in hemoglobinopathy drug development despite its potential. Adopting the ICH M15 credibility assessment framework and a question-centric approach can standardize regulatory interactions, increase transparency, and facilitate broader acceptance of M&S as evidence for benefit–risk decisions.

---



---

### ☒ Figures

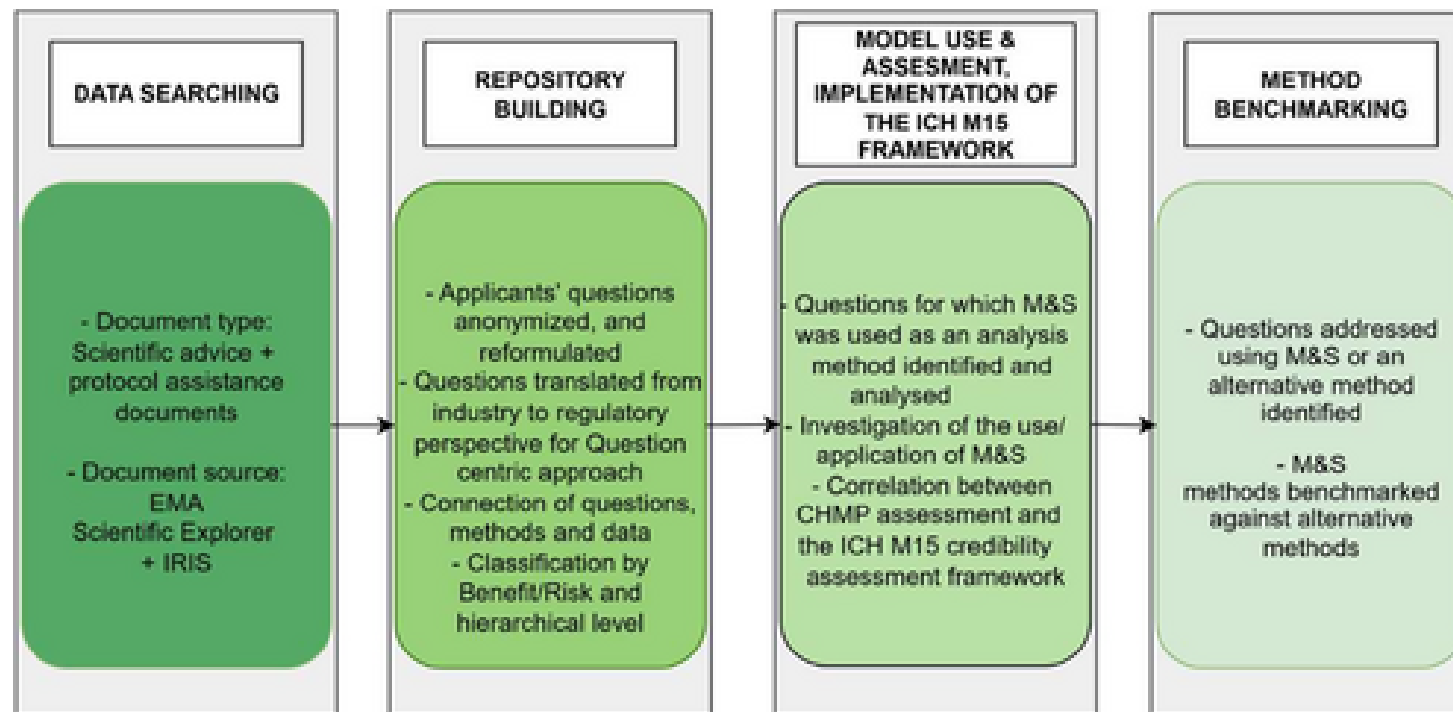


Figure 1: Schematic representation of the process used to compile the repository of regulatory questions, associated methods and data related to the benefit/risk balance o

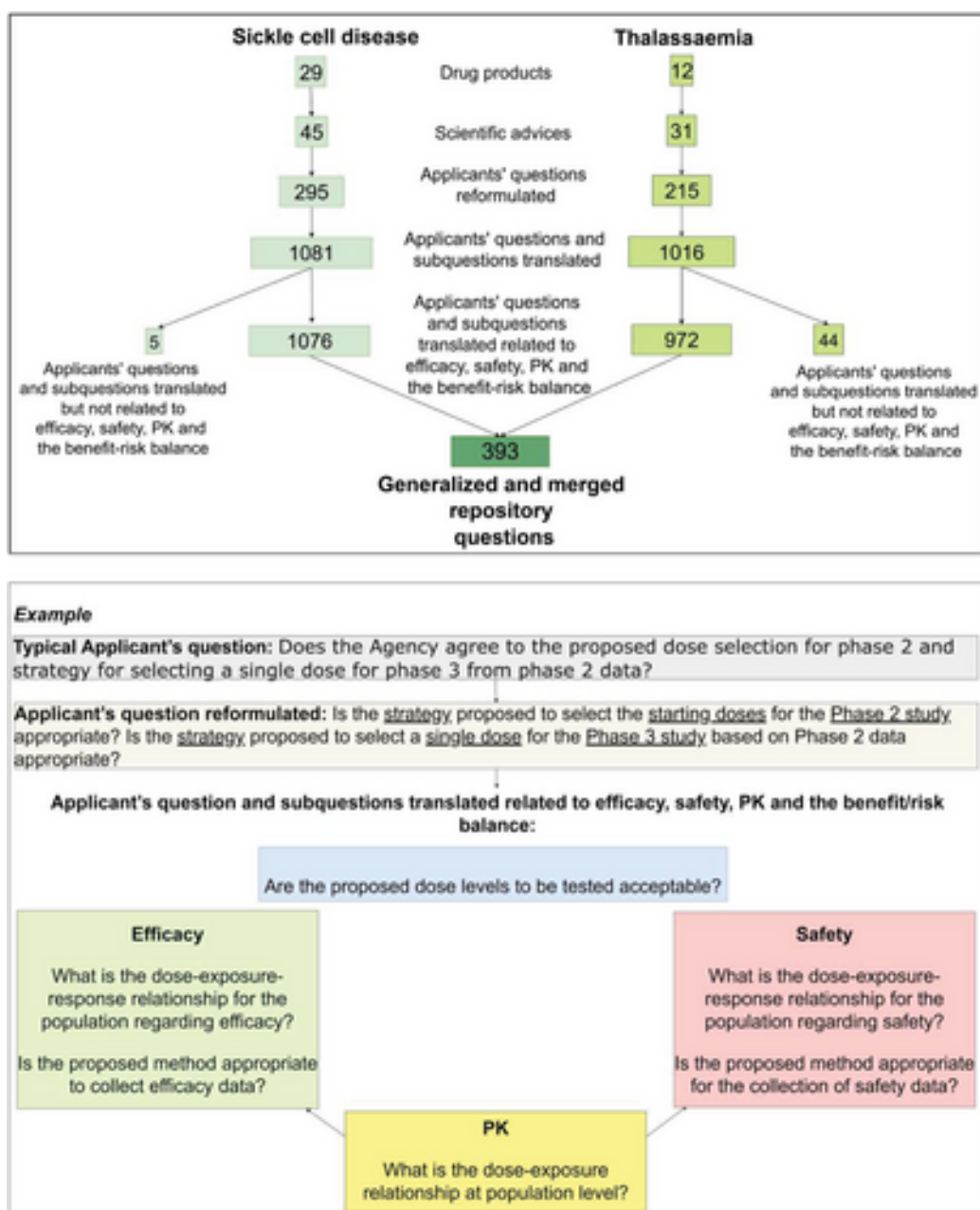


Figure 2: Steps taken to anonymize applicants' questions and reformulate them into repository questions. PK, pharmacokinetic.

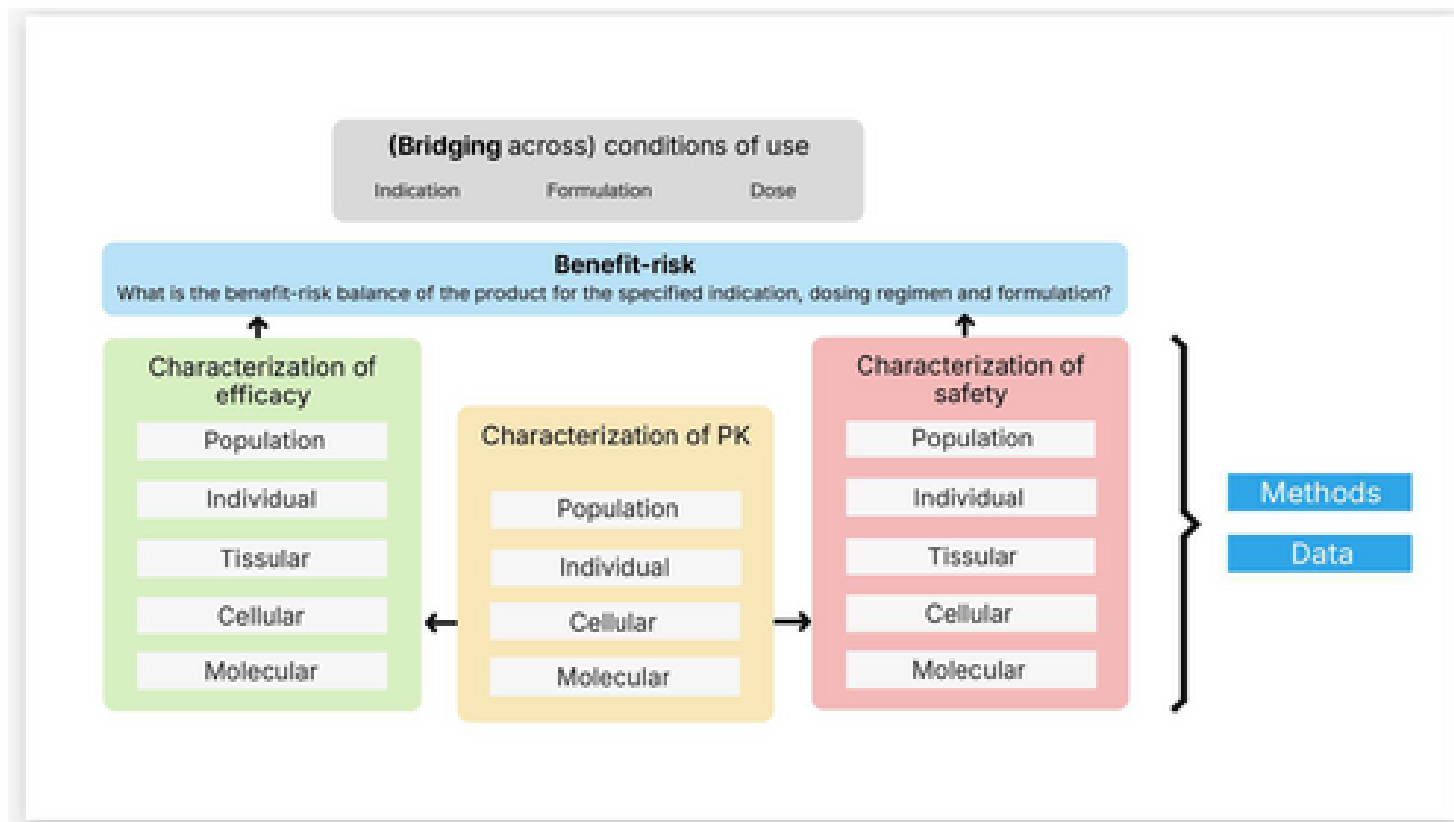


Figure 3: The overall structure of the repository. PK, pharmacokinetic.

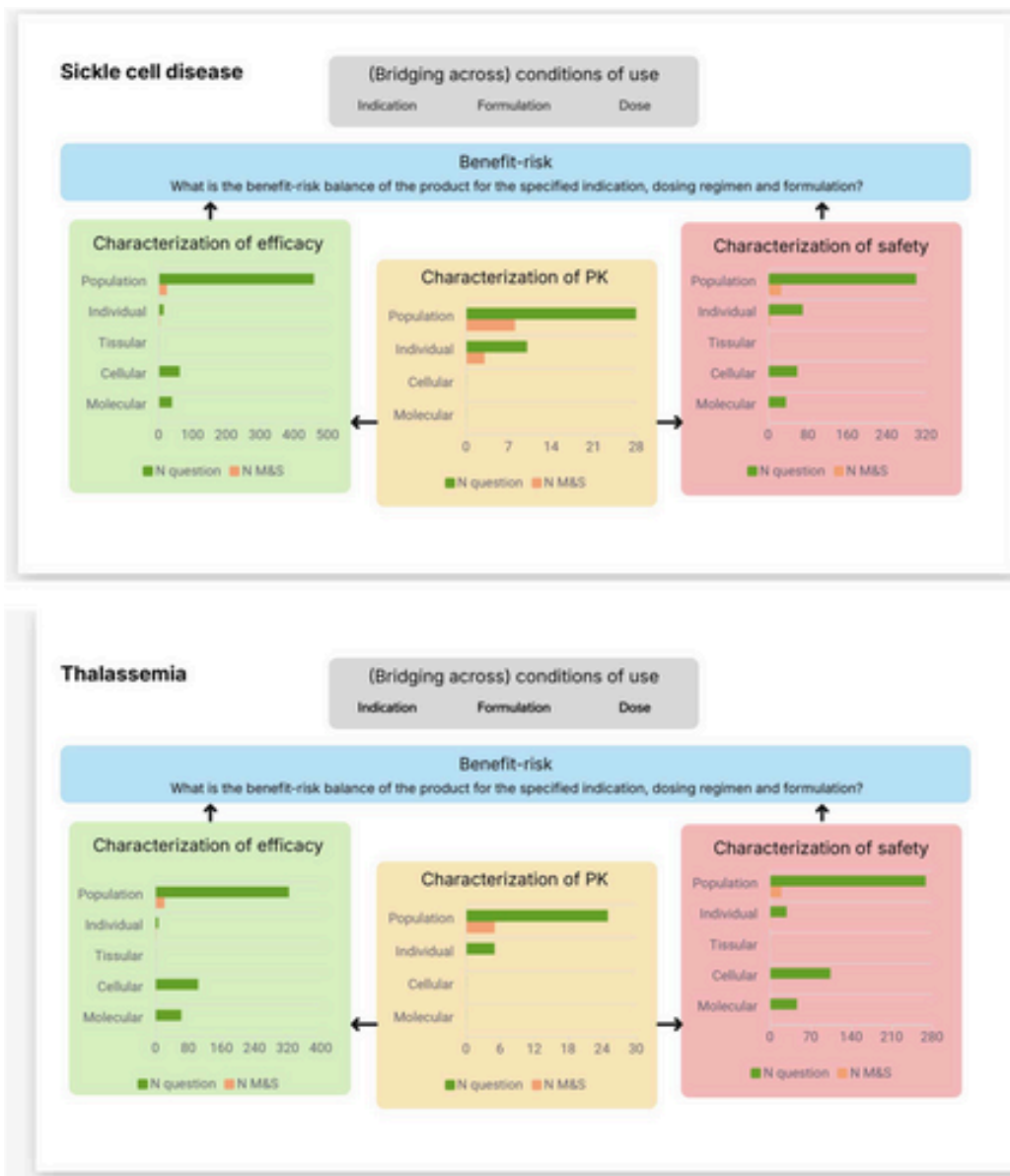


Figure 4: A detailed overview of the numbers of translated questions and sub-questions per benefit/risk classification and hierarchical level, including the questions addr

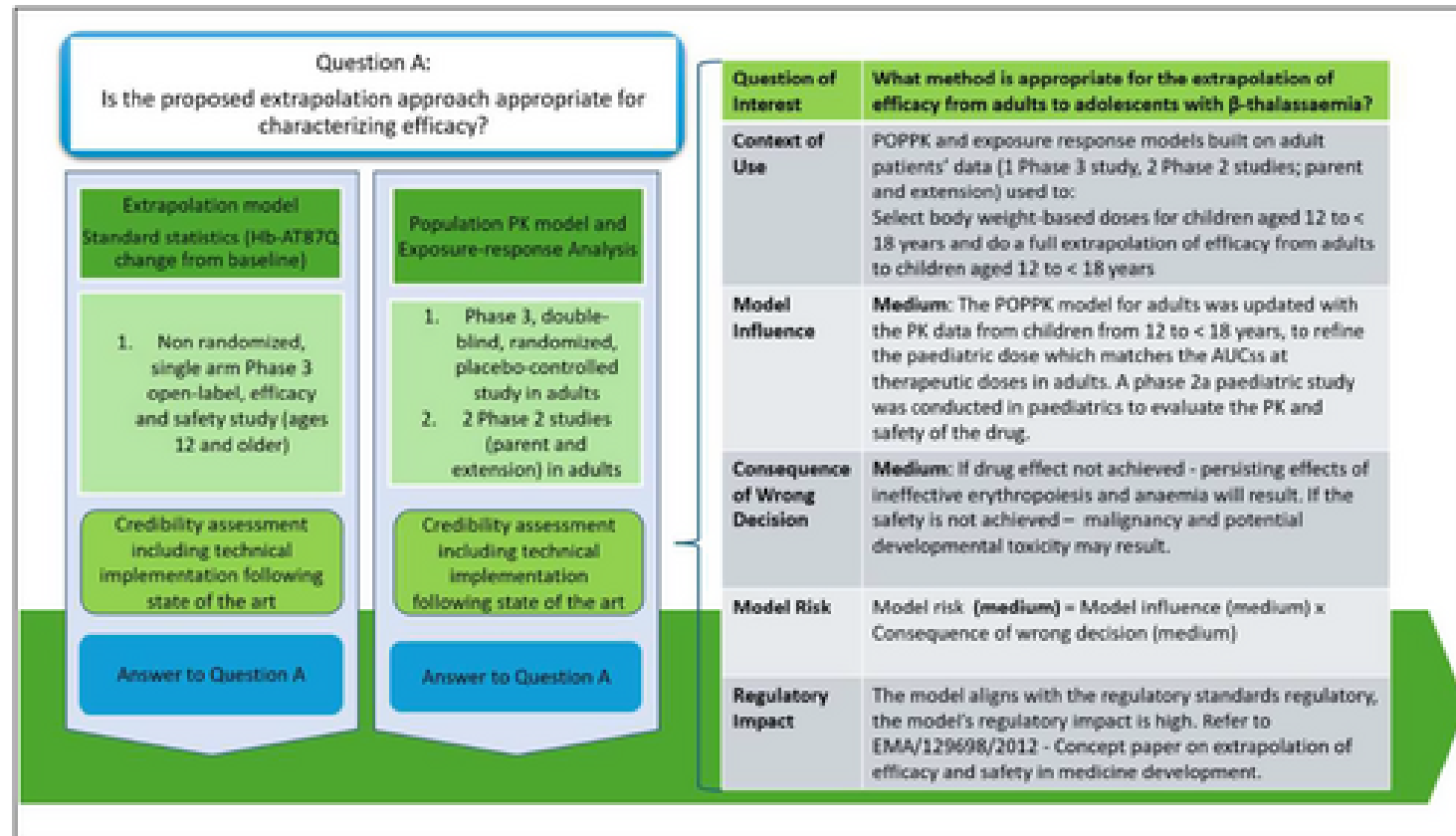


Figure 5: Credibility assessment of a model and benchmarking against alternative statistical analysis for addressing a drug development question related to extrapolation o